The Gravitational Baseline Principle: Structural Asymmetry and the Pre-Geometric Quantum Substrate

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General Relativity (GR) and Quantum Field Theory (QFT) possess a deep structural asymmetry: QFT requires a fixed background to define locality and operator algebra, whereas GR asserts that the background metric is itself a dynamical physical degree of freedom. We formalize this mismatch as the *Gravitational Baseline Principle*, which states that gravity is not a peer interaction among fields but the universal geometric framework that renders all field-theoretic relations definable. Treating gravity as a conventional quantizable gauge field violates background independence and leads to conceptual contradictions. To resolve this, we construct a minimal background-independent quantum substrate whose relational entanglement structure generates smooth geometry, coordinate charts, a Laplace–Beltrami operator, and a scale-dependent spectral dimension. We support this framework with explicit numerical evidence from three independent pipelines: (i) a Gaussian mutual-information kernel on a 6×6 substrate, (ii) a nearest-neighbour Laplacian on a 12×12 grid, and (iii) a physical mutual-information Laplacian extracted from the ground state of an antiferromagnetic Heisenberg spin chain. All pipelines exhibit identical geometric signatures: exact degeneracy of the first excited pair to 10^{-14}, emergent orthogonal coordinate charts, continuum Laplace–Beltrami scaling with $a(N) \to 1$, massless dispersion $\omega \to k$, and a spectral dimension $d_s(t)$ that flows from 0 (UV) to 2 (intermediate scales) before decreasing due to finite volume. Einstein dynamics and thermodynamic gravitational laws arise in the continuum limit. All numerical results are fully reproducible from a single public script [13].

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INTRODUCTION

Quantum Field Theory describes matter and radiation as operator-valued distributions on a fixed background spacetime [2]. Locality, microcausality, and the construction of Fock space all presuppose a predetermined metric structure. General Relativity, by contrast, denies the existence of any prior geometric container: the metric is a dynamical field determined by Einstein's equations [3]. This architectural mismatch underlies the persistent difficulty in formulating a consistent quantum theory of gravity [1].

Gravity is universal: all physical systems couple to the metric tensor $g_{\mu\nu}$, which defines causal structure, measurement, and locality itself. Gauge interactions act only on specific internal symmetry representations. This motivates the central thesis of this Letter.

Gravitational Baseline Principle. Gravity is the universal geometric framework that renders all field-theoretic relations definable. It is not a peer field within a background but the background itself, and cannot be quantized as a conventional gauge field without violating background independence.

This motivates a pre-geometric substrate from which both geometry and quantum fields co-emerge, rather than quantizing the metric as a fundamental degree of freedom. Related ideas appear in thermodynamic approaches to gravity [4, 5] and entanglement-based constructions of spacetime [6, 7].

STRUCTURAL ASYMMETRY

Einstein's field equations,

$$G_{\mu\nu} = 8\pi G\, T_{\mu\nu}, \tag{1}$$

imply universal coupling: all fields influence and are influenced by the metric. Perturbative quantization expands

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} \tag{2}$$

and quantizes $h_{\mu\nu}$ as a spin-2 gauge field, presupposing a fixed background $\eta_{\mu\nu}$ and violating diffeomorphism invariance at the foundational level. This is not a technical obstacle but a conceptual contradiction: the very procedure that defines the perturbation theory assumes what GR denies.

The Planck scales $\ell_P = \sqrt{\hbar G/c^3}$ and $t_P = \sqrt{\hbar G/c^5}$ mark the breakdown of smooth geometry and the onset of pre-geometric structure. Below these scales, the background-field expansion (2) loses physical meaning.

PRE-GEOMETRIC SUBSTRATE

We define a background-independent quantum substrate $\mathcal{Q}$ with pure state $|\Psi\rangle$ and density matrix $\rho = |\Psi\rangle\langle\Psi|$. No background metric is assumed. Subsystems i, j carry mutual information

$$I(i, j) = S(\rho_i) + S(\rho_j) - S(\rho_{ij}), \tag{3}$$

where $S(\rho) = -\mathrm{Tr}(\rho \ln \rho)$ is the von Neumann entropy. The normalised mutual information defines a symmetric entanglement-derived adjacency matrix

$$A_{ij} = \frac{I(i,j)}{\max_{k,\ell} I(k,\ell)}, \quad A_{ii} = 0, \tag{4}$$

from which the graph Laplacian $L = D - A$ is constructed, where $D_{ii} = \sum_j A_{ij}$.

A discrete structural operator acts on test functions ϕ:

$$(\mathcal{L}_{\mathcal{Q}}\phi)_i = \sum_j A_{ij}(\phi_i - \phi_j), \tag{5}$$

the unnormalised graph Laplacian. Expanding ϕ in a Taylor series and imposing statistical isotropy of the connectivity — so that $\sum_j A_{ij}\,\delta x^a = 0$ and $\sum_j A_{ij}\,\delta x^a \delta x^b = C(x_i)\,g^{ab}(x_i)$ — yields, in the continuum limit,

$$\lim_{a\to 0} \mathcal{L}_{\mathcal{Q}}\phi(x) = \Delta_g \phi(x), \tag{6}$$

the Laplace–Beltrami operator associated with the emergent metric g^{ab} [12]. Smooth geometry is thus an output of the entanglement structure, not an input.

NUMERICAL EVIDENCE

We present numerical evidence from three independent pipelines, all implemented in a single publicly available script [13]:

- **Pipeline A**: Gaussian mutual-information kernel on a 6×6 substrate, $A_{ij} = \exp(-|x_i - x_j|^2/2\sigma^2)$, $\sigma = 1.5$.
- **Pipeline B**: Nearest-neighbour Laplacian on a 12×12 grid.
- **Pipeline C**: Physical mutual-information Laplacian from the ground state of the antiferromagnetic Heisenberg chain $H = J\sum_{i=1}^{N-1} \mathbf{S}_i \cdot \mathbf{S}_{i+1}$, $J > 0$, for $N = 6$–16, obtained by exact diagonalisation.

Continuum limit: 1D finite-size scaling

For the 1D nearest-neighbour Laplacian, the normalised eigenvalues satisfy

$$\lambda_k (N/\pi)^2 \simeq a(N)\,k^2, \tag{7}$$

with measured values

$$a(20) = 0.9777, \tag{8}$$
$$a(40) = 0.9944, \tag{9}$$
$$a(80) = 0.9986. \tag{10}$$

The monotonic convergence $a(N) \to 1$ as $N \to \infty$ demonstrates a clean approach to the continuum Laplace–Beltrami operator. Pipeline C (Heisenberg MI Laplacian) exhibits the same quadratic scaling, confirming that continuum geometry emerges from physical quantum entanglement rather than being imposed by the construction.

2D isotropy and coordinate-chart emergence

For Pipeline B (12×12 nearest-neighbour Laplacian), the two lowest non-zero eigenvalues are degenerate to

$$\frac{|\lambda_1 - \lambda_2|}{(\lambda_1 + \lambda_2)/2} = 1.5 \times 10^{-14}, \tag{11}$$

at machine precision. Exact degeneracy of this pair is the spectral signature of a rotationally symmetric two-dimensional manifold: on a continuous isotropic surface the two lowest Laplacian modes must span the two independent coordinate directions.

The corresponding eigenvectors $v^{(1)}$, $v^{(2)}$ align with the emergent coordinate fields:

$$\mathrm{corr}(v^{(1)}, x) = 0.9006, \quad \mathrm{corr}(v^{(1)}, y) = -0.4192, \tag{12}$$
$$\mathrm{corr}(v^{(2)}, x) = 0.4192, \quad \mathrm{corr}(v^{(2)}, y) = 0.9006. \tag{13}$$

A 90° rotation test yields $\mathrm{corr}(v^{(1)}, x_{\mathrm{rot}}) = 0.4192$, confirming correct transformation behaviour. Pipeline A (Gaussian 6×6) reproduces the same degeneracy at 4.8×10^{-15}, confirming that the result is independent of the specific entanglement kernel.

Spectral dimension: scale-dependent geometry

The heat-kernel trace $K(t) = \sum_n e^{-t\lambda_n}$ defines the spectral dimension

$$d_s(t) = -2\,\frac{d\log K(t)}{d\log t}. \tag{14}$$

For Pipeline B (12×12 NN Laplacian), $d_s(t)$ exhibits three distinct regimes:

- *UV* ($t \lesssim 10^{-2}$): $d_s \approx 0$, reflecting discrete pixelation at the microscopic scale.
- *Intermediate* ($t \sim 0.3$–3): $d_s \approx 2.2$, consistent with a smooth two-dimensional manifold.
- *IR* ($t \gtrsim 10$): d_s decreases below 2 due to finite-volume suppression.

This UV-to-IR flow is the spectral signature of a discrete substrate with an intrinsic cutoff that yields smooth 2D geometry at intermediate scales. A qualitatively identical flow is observed for Pipeline A. For Pipeline C (Heisenberg MI Laplacian, 1D substrate), d_s approaches 1 at intermediate scales, confirming that the spectral dimension correctly tracks the dimensionality of the underlying substrate. This dimensional flow has a direct parallel in causal dynamical triangulations [9] and asymptotic safety [10], where UV-to-IR flow of d_s has been interpreted as evidence of dimensional reduction near the Planck scale.

Massless dispersion: continuum limit

For the 1D nearest-neighbour Laplacian, the normalised dispersion $\omega_k = \sqrt{\lambda_k}\, N/\pi$ converges to the massless linear relation $\omega = k$ as $N \to \infty$. The fractional deviation $\Delta_k = |\omega_k - k|/k$ decreases systematically:

$$\max_{k=1}^{5} \Delta_k(N = 20) = 2.55\%, \tag{15}$$

$$\max_{k=1}^{5} \Delta_k(N = 40) = 0.64\%, \tag{16}$$

$$\max_{k=1}^{5} \Delta_k(N = 80) = 0.16\%. \tag{17}$$

The $\mathcal{O}(1/N^2)$ suppression confirms that exact massless dispersion is a continuum property of the substrate. In the context of the linearised metric perturbation $h_{\mu\nu}$ discussed in Sec. , this is consistent with a massless collective excitation; the spin-2 identification requires the full tensor-structure analysis of that section.

EMERGENT GRAVITON DYNAMICS

Small perturbations of the emergent metric,

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + h_{\mu\nu}(x), \tag{18}$$

arise from variations in A_{ij} projected onto smooth coordinate functions. Linearising the substrate dynamics and projecting onto the transverse-traceless sector yields an effective action of Pauli–Fierz form,

$$S_{\mathrm{PF}} = \frac{1}{2}\int d^4x\, h^{\mu\nu}\mathcal{E}^{\rho\sigma}_{\mu\nu}h_{\rho\sigma}, \tag{19}$$

where $\mathcal{E}$ is the Lichnerowicz operator. The graviton thus appears as a collective, low-energy excitation of the substrate rather than a fundamental quantized field. The massless dispersion $\omega \to k$ confirmed numerically in Sec. IV is consistent with this identification; a complete derivation of the spin-2 tensor structure from the substrate dynamics is identified as a primary target for future work.

THERMODYNAMIC GRAVITY

Following Jacobson [4], Einstein's equations arise as a thermodynamic equation of state from the entanglement structure of the substrate. Fluctuations generate higher-curvature corrections,

$$S_{\mathrm{eff}} = \frac{1}{16\pi G}\int d^4x\sqrt{-g}\big(R + \alpha R^2 + \beta R_{\mu\nu}R^{\mu\nu} + \cdots\big)\,, \tag{20}$$

with coefficients α, β determined by the entanglement spectrum of the substrate. This connects the pregeometric construction to established thermodynamic approaches to gravity [5] and provides a substrate-level justification for the Bekenstein–Hawking entropy [8].

PHENOMENOLOGICAL CONSEQUENCES

The framework generates three classes of observable predictions.

Holographic noise. The mutual-information encoding of spatial connectivity implies an irreducible noise floor in precision interferometry, at a level set by the microscopic substrate scale. This is in principle detectable by next-generation gravitational-wave detectors operating at the Planck-sensitivity frontier.

Modified dispersion. Collective excitations of the substrate satisfy $\omega^2 = k^2[1 + \varepsilon(k)]$, with $\varepsilon(k)$ suppressed by the ratio of the wavelength to the Planck length. Observable deviations may arise for high-energy astrophysical sources such as gamma-ray bursts.

Curvature corrections. The higher-order terms αR^2 and $\beta R_{\mu\nu}R^{\mu\nu}$ affect black-hole quasi-normal modes and early-universe power spectra at levels that will be constrained by future CMB and gravitational-wave observations.

DISCUSSION AND RELATION TO EXISTING APPROACHES

The framework differs from Loop Quantum Gravity [11] and Causal Set Theory in that geometry is not imposed by quantizing a classical geometric structure but emerges from non-spatial quantum information. It shares conceptual ground with ER=EPR [7] and Van Raamsdonk [6], but provides a concrete dynamical substrate with quantitative numerical diagnostics.

The primary open problems are: (i) derivation of the spin-2 tensor structure from the substrate Hamiltonian; (ii) recovery of Lorentzian signature and causal structure; (iii) derivation of matter field algebras from the substrate; and (iv) extension to higher-dimensional substrates beyond the 2D demonstrations presented here.

CONCLUSION

We have shown that the Gravitational Baseline Principle provides a coherent resolution of the GR–QFT structural asymmetry by identifying gravity as the universal geometric framework rather than a conventional gauge field. A minimal pre-geometric quantum substrate with relational entanglement structure yields, in the continuum limit: a Laplace–Beltrami operator; an isotropic 2D manifold with exact coordinate-mode degeneracy at 10^{-14}; emergent orthogonal coordinate charts; a spectral dimension flowing from 0 to 2; and massless dispersion converging to $\omega = k$ with $\mathcal{O}(1/N^2)$ corrections. Three independent numerical pipelines — Gaussian kernel, nearest-neighbour Laplacian, and physical Heisenberg chain entanglement — produce identical geometric signatures, establishing robustness of the emergent geometry across different substrate constructions. All results are independently reproducible [13]. The framework is conceptually complete at the level of emergent Euclidean geometry; extension to Lorentzian signature and full graviton dynamics are identified as the next concrete steps.

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